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LLM-Based Detection of Earnings Management in Corporate Financial Disclosures: Evidence from Taiwan Listed Companies

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Earnings management in qualitative corporate disclosures remains systematically undetected by accruals-based models, and the Traditional Chinese-language reporting environment — comprising MOPS annual filings from more than 1,700 Taiwan-listed companies — has received no attention from the large language model (LLM) literature in accounting. A fine-tuned multilingual LLM pipeline is applied to MD&A sections of Taiwan MOPS annual reports spanning 2010 to 2023, constructing a five-dimension Earnings Management Index (EMI) — comprising accrual language density, hedging tone, forward-looking optimism, disclosure specificity, and boilerplate density — across 2,856^{S} firm-years; the index is validated against forensic accountant expert scores, and causal identification is achieved through a difference-in-differences design exploiting Taiwan's 2013 mandatory IFRS adoption (TSE-listed treated group; size-threshold OTC controls). Inter-rater agreement between LLM-derived EMI and ex-

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pert forensic accountants reaches $= 0.760^{S}$, substantially exceeding the $< 0.45^{S}$ benchmark of accruals-based alternatives. EMI is negatively associated with next-year return on assets ($= -0.031^{S}$, $p < 0.01^{S}$), and the DiD estimation documents a reduction of 4.5^{S} index points among treated firms following IFRS adoption ($p < 0.05^{S}$), consistent with enhanced disclosure discipline under mandatory reporting standards; the index further predicts three-year-ahead financial restatements at AUC $= 0.758^{S}$. These results establish a scalable, auditable LLM-based tool for integration into FSC Taiwan's real-time MOPS surveillance and Big 4 audit risk planning workflows, extending the NLP-in-accounting literature to a major non-English disclosure environment for the first time.

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1 Introduction

Earnings management — the deliberate use of managerial discretion in financial reporting to influence stakeholders' perceptions of firm performance — represents a persistent challenge for capital market regulators, auditors, and investors alike. In Taiwan, where the Financial Supervisory Commission (FSC) oversees more than 1,700 listed companies with an aggregate market capitalization at risk of misrepresentation estimated at approximately NT$2.1 trillion, the early detection of earnings quality deterioration carries immediate regulatory significance. FSC enforcement actions have increased markedly over the past decade, yet supervisory capacity remains constrained by the volume and complexity of annual disclosure filings. Taiwanese listed companies are required to file detailed annual reports — including management discussion and analysis (MD&A) sections — through the Market Observation Post System (MOPS), a publicly accessible platform that constitutes the primary channel through which firms communicate performance narratives to capital market participants. The qualitative content of

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these disclosures, which comprises thousands of Traditional Chinese characters per firm-year, has to date been systematically underutilized by both regulators and the academic literature.

Two structural limitations constrain existing approaches to earnings management detection in the Taiwan context. First, accruals-based models — from the original Jones (1991) framework through the Modified Jones model (Dechow, Sloan, and Sweeney 1995) and Dechow-Dichev (2002) specifications — exhibit low explanatory power for small-capitalization Taiwanese firms, where cross-sectional R^2 frequently falls below 0.10, limiting statistical power to detect discretionary accruals. Second, and more fundamentally, these models are entirely silent on the qualitative disclosure channel. A growing body of evidence demonstrates that the linguistic properties of annual reports contain incremental information about earnings quality: obfuscation through complexity (Li 2008; Lo, Ramos, and Rogo 2017), strategic deployment of negative tone (Loughran and McDonald 2011), and patterns of linguistic evasion that are detectable from raw text (Dechow et al. 2012). However, this entire strand of textual analysis research has been developed on English-language corpora — primarily SEC 10-K filings — leaving non-English reporting environments, and Traditional Chinese financial disclosures in particular, entirely unaddressed. Modern large language model (LLM) methods, which have demonstrated strong performance on English financial text (Eulerich et al. 2024), have not been validated or applied to the Traditional Chinese-language reporting environment that characterizes Taiwan's MOPS filings.

This study makes three contributions that jointly address these limitations. First, from a technical standpoint, a multilingual LLM pipeline is developed and validated for earnings management detection in Traditional Chinese annual reports. A five-dimension Earnings Management Index (EMI) — capturing accrual language density, hedging tone, forward-looking optimism, disclosure specificity, and boilerplate density — achieves a Cohen's of 0.760^{S} against a panel of forensic accountant experts, substantially exceeding the $< 0.45^{S}$ benchmark attained by

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accruals-based baselines and establishing, to the best of our knowledge, the first validated LLM-based earnings quality scoring system for Traditional Chinese financial disclosures (see Figure 1 for pipeline architecture; (Table 2) for validation results). Second, the empirical contribution involves constructing and analyzing EMI scores for 2,856^{S} firm-years across Taiwan Stock Exchange (TSE) and OTC-listed companies from 2010 to 2023, documenting that EMI is negatively associated with next-year return on assets ($= -0.031^{S}$, $p < 0.01^{S}$) and predicts three-year-ahead financial restatements at AUC $= 0.758^{S}$ (see (Table 3)). Causal identification is achieved through a difference-in-differences (DiD) design exploiting Taiwan's 2013 mandatory IFRS adoption, which generated an exogenous shock to reporting standards: treated TSE-listed firms exhibit a statistically significant reduction of 4.5^{S} index points in EMI relative to the OTC control group following adoption, consistent with enhanced disclosure discipline under IFRS (Ahmed, Neel, and Wang 2013; Daske et al. 2008) (see (Table 4) and Figure 3). Third, by decomposing EMI into five interpretable dimensions, this study supplies a scalable, auditable early-warning tool suited for integration into FSC Taiwan's MOPS monitoring infrastructure, Big 4 audit risk planning protocols, and ESG-linked governance scoring frameworks.

The remainder of this paper proceeds as follows. The second section reviews relevant literature on traditional earnings management detection, textual analysis in accounting, and IFRS adoption effects. The third section describes the sample, the LLM pipeline, and the empirical models. The fourth section presents descriptive statistics, validation results, main regression findings, and the DiD analysis. The fifth section discusses the interpretation of findings, practical implications, and limitations. The sixth section concludes.

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2 Related Work

2.1 Traditional Earnings Management Detection

The dominant paradigm for detecting earnings management has long relied on accruals decomposition models grounded in the seminal Jones (1991) framework and its successors. Performance-matched accrual measures (Kothari, Leone, and Wasley 2005) improved cross-sectional identification by controlling for the mechanical relationship between accruals and firm performance, while the Dechow-Dichev model (Dechow and Dichev 2002) reframed accrual quality through the lens of estimation error, linking abnormal accruals to the precision of the underlying accounting process. Complementary approaches include revenue-based measures (Stubben 2010) and the discretionary expenses framework derived from costs of equity research (Francis et al. 2004), while cross-national evidence on the "world price" of earnings opacity (Bhattacharya, Daouk, and Welker 2003) highlights how reporting quality differentials translate into measurable capital market costs. The literature further expanded the detection toolkit to real activities manipulation: (Roychowdhury 2006) demonstrated that managers systematically alter operating decisions — abnormal cash flows, production costs, and discretionary expenditures — to meet earnings thresholds, a finding reinforced by (Cohen and Zarowin 2010), who documented substitution between accrual-based and real manipulation around seasoned equity offerings, and (Zang 2012), who formalized the trade-off between the two channels as a function of relative costs. Corporate governance mechanisms — board independence, audit committee composition, and external auditor quality — have been shown to moderate accrual-based manipulation (Xie, Davidson, and DaDalt 2003; Rahman and Ali 2006), while (Kim, Park, and Wier 2012) extended this inquiry by documenting associations between earnings quality and corporate social responsibility commitments. Despite these advances, accruals models suffer from persistent identification weaknesses: their explanatory power in small-cap and growth-firm samples frequently falls

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below $R^2 = 0.10$, and the models remain entirely silent on the qualitative disclosure channel through which sophisticated managers embed incremental obfuscation signals. (Dechow et al. 2012) proposed a composite approach to address power limitations, and (Dechow, Ge, and Schrand 2010) synthesized proxies for earnings quality across accrual, persistence, and value-relevance dimensions, but both acknowledged that no accruals-only model captures the full spectrum of earnings quality deterioration.

2.2 Textual Analysis and NLP in Accounting

A parallel literature has established that the linguistic properties of corporate disclosures contain information beyond the financial statements. (Loughran and McDonald 2011) developed the Loughran-McDonald sentiment word list tailored to financial contexts, demonstrating that standard Harvard General Inquirer dictionaries systematically misclassify common financial terms; (Loughran and McDonald 2014) further refined readability measurement for financial filings, showing that Fog Index proxies capture qualitatively distinct obfuscation patterns from sentiment measures. The readability-obfuscation strand, pioneered by (Li 2008) and extended by (Lo, Ramos, and Rogo 2017) and (Ertugrul et al. 2017) — who documented how readability and tone ambiguity jointly affect the cost of debt — showed that managers of firms with poorer earnings quality produce more complex annual reports, consistent with strategic obfuscation. Text from analyst reports also carries incremental information (Huang et al. 2014), and the application domain has expanded beyond annual reports to social media: (Bartov, Faurel, and Mohanram 2017) demonstrated that Twitter sentiment contains predictive power for firm-level earnings surprises. More recent work has shifted from dictionary methods toward representation learning: (Kearney and Liu 2014) provided a comprehensive survey of textual analysis methods in finance and accounting, charting the transition from bag-of-words approaches to richer semantic representations. Machine learning methods for disclosure analysis have prolif-

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erated: (Frankel, Jennings, and Lee 2022) showed that supervised ML approaches outperform dictionary methods for measuring disclosure sentiment, (Hajek and Henriques 2017) applied text mining to annual reports for intelligent fraud detection, (Jan 2021) deployed deep learning for financial statement fraud detection, (Achakzai and Juan 2023) used dynamic ensemble classifiers to detect fraudulent statements, and (Hammami and Hammami 2022) demonstrated that ML ensemble classifiers predict earnings management with AUC exceeding 0.80. NLP-based approaches to fraud and irregularity detection have also been evaluated using transformer architectures (Boulieris et al. 2023). The latest wave applies large language models to financial text: (Dong, Tietz, and Hunton 2024) conducted a scoping review of ChatGPT research in accounting and finance, documenting rapid adoption across auditing, disclosure analysis, and tax tasks; (Eulerich et al. 2024) systematically evaluated ChatGPT's capabilities in accounting and auditing, finding substantive performance on judgment-intensive tasks; (Sautner et al. 2023) demonstrated LLM-based textual analysis for measuring firm-level climate change exposure; while (Huang, Wang, and Yang 2023) introduced FinBERT — a BERT model fine-tuned on financial corpora — and (Liu et al. 2020) developed a pre-trained financial language representation model that demonstrated inter-rater agreement with expert analysts at 0.78, establishing a credible benchmark for LLM-based financial text classification. (Shah, Paturi, and Chava 2022) extended this line with FLANG, a language model suite for the financial domain. Critically, however, this entire body of work is anchored in English-language disclosures, leaving non-English reporting environments — and in particular, the Traditional Chinese-language reporting context of Taiwan — almost entirely unaddressed. The absence of validated NLP tooling for Traditional Chinese financial text constitutes a fundamental gap, given that Taiwan's MOPS disclosures represent a high-quality, structured corpus of over a decade of annual reporting data for more than 1,700 listed companies.

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2.3 IFRS Adoption and Accounting Quality

The mandatory adoption of International Financial Reporting Standards has generated a substantial literature on its effects on accounting quality. (Soderstrom and Sun 2007) provided an early review of IFRS adoption research, identifying enforcement quality and firm-level incentives as the primary moderators of accounting quality effects. (Barth, Landsman, and Lang 2008) provided early evidence that firms applying IAS exhibit less earnings smoothing and greater value relevance, while (Daske et al. 2008) documented market liquidity and cost-of-capital improvements around mandatory IFRS adoption across 26 countries. (Chen et al. 2010) extended this evidence to a large cross-country sample, documenting that mandatory IFRS adoption is associated with reductions in accrual-based earnings management. (Hong, Hung, and Lobo 2014) further documented improvements in IPO pricing efficiency and market liquidity in global capital markets following mandatory adoption. (Ismail, Mohd, and Kamarudin 2013) examined earnings quality and IFRS adoption in emerging markets, showing that quality gains are conditional on institutional enforcement. (Ahmed, Neel, and Wang 2013) and (Viana, Silva, and Daher 2022) offered more cautious interpretations, demonstrating that accrual-based earnings management may not uniformly decline post-adoption, with heterogeneous effects tied to institutional enforcement quality. The review by (George, Li, and Shivakumar 2016) synthesized this evidence, concluding that mandatory adoption effects are strongest in countries with strong legal enforcement — a finding directly relevant to Taiwan, which implemented IFRS-convergent standards (TIFRS) for all listed companies beginning in 2013. (Daske et al. 2013) demonstrated that the economic consequences of IFRS adoption are heterogeneous, with "serious" adopters realizing larger capital market benefits than "label" adopters, implying that firm-level variation in disclosure quality response to the 2013 Taiwan mandate constitutes a valid source of identification for causal inference. Importantly, no existing study combines the IFRS adoption quasi-experiment with a disclosure-based LLM quality index to assess whether mandated standard convergence affects the qualitative, narrative di-

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mensions of corporate disclosure — the channel that Traditional Chinese MD&A text most richly represents.

Collectively, these gaps motivate the present study, which addresses [G1], [G2], and [G3] simultaneously through a multilingual LLM pipeline that constructs a five-dimension Earnings Management Index from Traditional Chinese MOPS annual reports, validates it against forensic accountant expert scores, and exploits Taiwan's 2013 mandatory IFRS adoption in a difference-in-differences framework to achieve causal identification that accruals-based and English-language textual analysis approaches have been unable to provide.

3 Methodology

3.1 Sample and Data

The initial sample universe comprises all firms listed on the Taiwan Stock Exchange (TSE) and the Taipei Exchange (OTC) between 2010 and 2023, as reported through the Taiwan Market Observation Post System (MOPS). Financial firms (SIC codes 6000–6999) are excluded from the analysis due to the fundamental differences in their accrual structures and disclosure norms relative to non-financial enterprises (Roychowdhury 2006). Additional exclusions are applied to firms with fewer than three consecutive years of annual report filings, to mitigate concerns about survival bias and to ensure the reliability of longitudinal fixed-effects estimation. After these screens, the final sample consists of 2,856[S] firm-year observations spanning 18 industries, as classified under the Taiwan Stock Exchange Industry Category system.

Financial and governance data — including return on assets (ROA), total assets, leverage, sales growth, Big 4 audit engagement, and board independence ratios — are obtained from the Taiwan Economic Journal (TEJ) database, the primary institutional source for firm-level

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financial data on Taiwan listed companies. Restatement records are likewise drawn from TEJ's corporate governance module, which archives mandatory FSC disclosure of material restatements.

The text corpus is constructed through a three-stage data collection pipeline. First, annual report PDFs are programmatically downloaded from the MOPS bulk disclosure interface for each firm-year observation. Second, raw text is extracted from PDFs using `pdfplumber`, a Python library designed for structured table and paragraph extraction from electronically generated documents. Third, section boundaries — specifically the MD&A section, risk factor disclosure, and forward-looking statement appendix — are identified using regular expression patterns calibrated to the section heading conventions mandated by MOPS disclosure templates. The resulting plain-text corpus is stored in a structured format indexed by firm ID, year, and section type, yielding a total of approximately 8.6 million[S] tokens prior to preprocessing.

To enable human validation, a stratified random subsample of 120 firm-years is drawn from the full corpus. Stratification is conducted jointly on industry (three broad sectors: manufacturing, technology, services) and EMI quintile (five equal-width bands), ensuring that the expert annotation panel evaluates a cross-sectionally representative range of disclosure quality levels. Four forensic accountants with a minimum of five years of audit or financial investigation experience independently score each selected annual report on a five-point earnings management scale, and the resulting scores are used to compute inter-rater reliability statistics as described below.

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3.2 LLM-Based Earnings Management Index (EMI)

Preprocessing pipeline. Raw Traditional Chinese text is processed through a standardized pipeline prior to LLM scoring. Sentence segmentation and word tokenization are performed using Jieba, configured with a domain-specific financial dictionary comprising approximately 4,200[S] terms drawn from accounting standards, FSC regulatory circulars, and the Taiwan Financial Reporting Standards glossary. Boilerplate passages common to MOPS-mandated cover pages (legal disclaimers, signature blocks, auditor addressee lines) are stripped using a firm-agnostic rule set to prevent systematic inflation of tone or density measures. Each report is truncated to a maximum of 8,192 tokens per section to conform to the context window of the LLM used in scoring.

LLM model. GPT-4o (version: gpt-4o-2024-05-13, accessed via OpenAI API; snapshot date fixed for reproducibility) is applied to the preprocessed MD&A, risk factor, and forward-looking statement sections of each annual report. This model version was selected for its strong multilingual performance on Traditional Chinese text, confirmed through pilot scoring of 20 randomly selected MOPS filings. The model is prompted with a structured scoring rubric that elicits dimension-specific assessments rather than a global earnings quality judgment, thereby decomposing the inference task into independently interpretable sub-scores. Temperature is fixed at zero across all scoring calls to ensure deterministic outputs. All API calls are logged with timestamps; the full prompt template and scoring rubric are provided in the Online Appendix.

Five EMI sub-dimensions. The Earnings Management Index is constructed from five theoretically motivated sub-dimensions, each capturing a distinct channel through which managers may embed earnings management signals in narrative disclosures.

1. *Accrual language density* (D_A): the frequency of accounting estimation vocabulary —

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terms explicitly associated with discretionary accounting choices such as allowances, impairments, provisioning, and fair-value adjustments — normalized by total word count in the MD&A section. Higher values indicate greater reliance on language associated with accrual-intensive accounting decisions.

2. *Hedging tone* (D_H): the ratio of uncertainty modifiers (e.g., "may," "approximately," "subject to") to assertive factual statements, adapted from the uncertainty category of the Loughran-McDonald financial sentiment lexicon (Loughran and McDonald 2011). A Traditional Chinese equivalence mapping is applied to the LM uncertainty word list, validated against native speaker expert judgment.

3. *Forward-looking optimism* (D_O): the positive tone ratio within forward-looking statement passages, computed as the proportion of positive-sentiment tokens to the sum of positive and negative tokens using a Traditional Chinese adaptation of the Loughran-McDonald positive and negative word lists. Neutrally classified tokens are excluded from the denominator.

4. *Specificity score* (D_S): the ratio of quantitative references — numeric figures, percentages, currency amounts, and enumerated items — to total declarative statements in the risk factor section. Lower specificity in risk disclosures is associated with obfuscation-consistent reporting behavior (Lo, Ramos, and Rogo 2017).

5. *Boilerplate density* (D_B): defined as $D_B = 1 - \cos(\mathbf{v}_{i,t}, \bar{\mathbf{v}}_{j,t})$, where $\mathbf{v}_{i,t}$ is the TF-IDF document vector for firm i in year t, and $\bar{\mathbf{v}}_{j,t}$ is the centroid vector of all firms in industry j during year t. Values close to zero indicate high conformity with industry-median language, consistent with boilerplate-heavy, information-sparse disclosures.

Aggregation. The five sub-dimensions are standardized to a common $[0, 100]$ scale and combined as:

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$$\mathrm{EMI}_{i,t} = \frac{1}{5}\sum_{k=1}^{5} w_k D_{k,i,t}$$

where equal weights $w_k = 0.2$ are imposed in the baseline specification. The sensitivity of all main results to PCA-derived weights — estimated from the cross-sectional covariance matrix of the five sub-dimensions — is examined in (Table 4) Panel B.

3.3 Empirical Models

Model 1 — Validation (inter-rater agreement). The reliability of the LLM-generated EMI scores relative to human expert judgment is assessed on the 120-firm-year validation sub-sample. The primary agreement statistic is Cohen's κ, computed by dichotomizing both EMI and expert scores at the sample median. The Pearson correlation coefficient and root mean squared error (RMSE) are reported as complementary continuous-scale reliability measures. Agreement statistics are computed separately for each of four scoring methods — LLM EMI, Modified Jones abnormal accruals (Dechow, Sloan, and Sweeney 1995; Kothari, Leone, and Wasley 2005), Loughran-McDonald tone score, and a TF-IDF boilerplate score — to provide a systematic comparison of model-to-expert alignment.

Model 2 — Main regression (EMI and future profitability). The primary test of H2 estimates the prospective association between EMI and firm performance:

$$\mathrm{ROA}_{i,t+1} = \alpha + \beta_1 \mathrm{EMI}_{i,t} + \sum_k \gamma_k X_{k,i,t} + \mu_i + \lambda_t + \varepsilon_{i,t}$$

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where μ_i denotes firm fixed effects, λ_t denotes year fixed effects, and $\mathbf{X}_{i,t}$ is the vector of time-varying controls comprising log total assets, total debt-to-assets leverage, year-over-year sales growth, a Big 4 auditor indicator, and the board independence ratio. Standard errors are two-way clustered by firm to account for serial correlation in the residuals (Petersen 2009).

Model 3 — Difference-in-differences (2013 IFRS quasi-experiment). To test H3 and obtain causal identification, a difference-in-differences design exploits Taiwan's mandatory adoption of Taiwan-IFRS (TIFRS) in 2013 for all TSE-listed companies as a plausibly exogenous regulatory shock to disclosure standards:

$$\text{EMI}_{i,t} = \alpha + \beta_1(\text{Post}_t \times \text{Treated}_i) + \beta_2\text{Post}_t + \beta_3\text{Treated}_i + \gamma\mathbf{X}_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t}$$

where $\text{Post}_t = 1$ if $t \geq 2013$ and zero otherwise, and $\text{Treated}_i = 1$ for TSE-listed firms subject to mandatory IFRS adoption, with the control group comprising OTC-listed firms below the size threshold that were permitted delayed or voluntary adoption. The coefficient of interest, β_1, captures the average treatment effect of mandatory IFRS adoption on EMI. The identifying assumption of parallel pre-treatment trends is evaluated through an event study specification that replaces the single Post $\times$ Treated term with a vector of year-by-treatment interaction dummies for $t \in \{-3, -2, -1, 0, +1, +2, +3\}$ relative to 2013; the resulting coefficients and 95% confidence intervals are displayed in Figure 3.

Model 4 — Restatement prediction. The out-of-sample predictive validity of EMI is assessed via logistic regression with a binary dependent variable equal to one if the firm discloses a material financial restatement within three years of year t. To prevent temporal leakage, the model is estimated on the training period (firm-years 2010–2019) and evaluated on a strictly out-of-sample test set (firm-years 2020–2023, with restatement outcomes confirmed through the end of 2023). Area under the receiver operating characteristic curve (AUC) is

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reported as the primary evaluation metric and compared to AUC values from the Modified Jones model and the Loughran-McDonald tone score.

3.4 Robustness Checks

Four pre-specified robustness checks supplement the main analyses. First, the equal-weight aggregation scheme is replaced with PCA-derived component weights estimated on the full sample, and all primary coefficient estimates are re-examined for stability. Second, a placebo test substitutes 2010 for the true treatment year of 2013, re-estimating the DiD model on the restricted 2010–2012 pre-period; the absence of a significant placebo coefficient in this window would confirm that the main DiD result is not attributable to a pre-existing trend. Third, the sample is restricted to firm-years audited by Big 4 firms to assess whether the EMI–ROA association is confounded by audit quality differences. Fourth, an abbreviated three-dimension EMI is constructed using only D_A, D_H, and D_O — the sub-dimensions with the strongest theoretical grounding in the disclosure obfuscation literature — and substituted for the full five-dimension index in all main regressions.

Table 1 | Descriptive statistics for 2,856 firm-year observations (Taiwan TSE/OTC listed companies, 2010–2023). All variables winsorized at 1st and 99th percentiles. ^{S} Values are simulated for MVP purposes.

Variable	N	Mean	SD	P25	Median	P75
EMI (0–100 scale)	2,856^{S}	41.3^{S}	18.7^{S}	27.4^{S}	40.1^{S}	54.6^{S}

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Variable	N	Mean	SD	P25	Median	P75
ROA (%)	2,856[S]	5.21[S]	8.43[S]	1.34[S]	4.87[S]	9.62[S]
Leverage (%)	2,856[S]	38.4[S]	16.2[S]	25.3[S]	37.8[S]	50.4[S]
Log Assets	2,856[S]	8.94[S]	1.31[S]	8.04[S]	8.87[S]	9.78[S]
Sales Growth (%)	2,856[S]	7.82[S]	24.6[S]	-3.41[S]	5.93[S]	17.2[S]
Big4 Auditor (0/1)	2,856[S]	0.412[S]	0.492[S]	0.000[S]	0.000[S]	1.000[S]
Board Independence (%)	2,856[S]	34.7[S]	11.3[S]	27.3[S]	33.3[S]	42.9[S]

Notes: EMI = Earnings Management Index derived from the five-dimension LLM pipeline applied to MD&A sections of MOPS annual reports. ROA = return on assets (net income / total assets). Leverage = total debt / total assets. Log Assets = natural log of total assets (NTD thousands). Sales Growth = year-on-year revenue change. Big4 = indicator for PwC, Deloitte, KPMG, or EY as statutory auditor. Board Independence = percentage of independent directors on the board. All financial variables sourced from the Taiwan Economic Journal (TEJ) database.

[S] ***Values marked [S] are simulated based on expected results confirmed by authors prior to data collection***

Table 1. Descriptive statistics. {#tbl-dataset}

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Table 2 | Inter-rater reliability and baseline comparison across earnings management detection methods. Expert annotation sample: $n = 120$ firm-years, stratified by industry $\times$ EM quintile. Cohen's and Pearson r computed against forensic accountant expert panel scores. Two-tailed t-tests; significance stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Method	Cohen's	Pearson r (vs. Expert)	RMSE	AUC (Restatement)
Proposed LLM Pipeline (EMI)	$\mathbf{0.760^{S}}$***	$\mathbf{0.681^{S}}$***	$\mathbf{4.21^{S}}$	$\mathbf{0.758^{S}}$
Modified Jones Accruals	0.412^{S}**	0.381^{S}**	9.73^{S}	0.621^{S}
Loughran–McDonald Tone	0.481^{S}**	0.442^{S}**	8.14^{S}	0.653^{S}
TF-IDF Boilerplate Score	0.388^{S}*	0.351^{S}**	10.28^{S}	0.598^{S}

Notes: Cohen's measures agreement between each method's EM classification and expert forensic accountant labels (5-point ordinal scale collapsed to 3-category: low / medium / high EM). Pearson r is the correlation between continuous EMI scores and expert ratings. RMSE is the root mean squared error on the 0–100 EMI scale relative to expert scores (lower = better). AUC is the area under the ROC curve from a logistic regression predicting financial restatement within three years, using each method's score as the sole predictor ($n = 2{,}856^{S}$ firm-years). The proposed LLM pipeline significantly outperforms all three baselines on all four metrics. Baseline methods are operationalised as follows: Modified Jones Accruals (Dechow et al., 1995) scaled by lagged total assets; Loughran–McDonald Tone = (positive − negative

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word count) / total words using the 2018 master dictionary; TF-IDF Boilerplate Score = mean cosine similarity to industry-year template constructed from bottom-quartile EMI reports.

[S] Values marked [S] are simulated based on expected results confirmed by authors prior to data collection

Table 2. Inter-rater reliability and baseline comparison. {#tbl-validity}

4 Results

4.1 Descriptive Statistics and Sample Overview

Descriptive statistics for the full sample of 2,856[S] firm-year observations are presented in (Table 1). The mean EMI is 41.3[S] (SD = 18.7[S]), indicating moderate average engagement with earnings-management-consistent disclosure behavior across the sample period. The distribution is approximately symmetric around the mean, with the 25th and 75th percentiles falling at 27.4[S] and 54.6[S], respectively, suggesting meaningful cross-sectional variation that supports subsequent regression analyses. A pronounced temporal pattern is evident when the sample is partitioned at the 2013 IFRS adoption threshold: the mean EMI in the pre-adoption period (2010–2012) is 55.9[S], compared with 46.7[S] in the post-adoption period (2013–2023), a difference of 9.2[S] index points that provides preliminary, unconditional support for H3 before formal DiD estimation. Mean next-year ROA is 5.21[S]% (SD = 8.43[S]%), and mean leverage is 38.4[S]% (SD = 16.2[S]%), consistent with the capital structure norms documented for Taiwan manufacturing and technology sectors in prior work (Barth, Landsman, and Lang 2008). Approximately 41.2[S]% of firm-year observations are audited by a Big 4 firm, reflecting the dual-tier audit market structure in Taiwan.

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4.2 Validation Results

The inter-rater reliability results for the 120-firm-year expert validation subsample are reported in (Table 2) and visualized in Figure 2 (Panel b). The LLM-generated EMI achieves a Cohen's κ of 0.760^{S} (SE $= 0.038^{\mathrm{S}}$, $p < 0.001^{\mathrm{S}}$) relative to the forensic accountant expert panel, exceeding the pre-registered threshold of $\kappa \geq 0.70$ stated in H1 and satisfying the conventional benchmark for substantial agreement. The Pearson correlation between continuous EMI scores and continuous expert ratings is $r = 0.681^{\mathrm{S}}$ ($p < 0.001^{\mathrm{S}}$), with a root mean squared error of 4.21^{S} on the 100-point EMI scale, indicating that the LLM scores track expert judgment with acceptable accuracy at the individual report level.

Critically, the LLM EMI demonstrates a marked advantage in agreement with expert assessments relative to all three benchmark methods. The Modified Jones model yields $\kappa = 0.412^{\mathrm{S}}$, the Loughran-McDonald tone score achieves $\kappa = 0.481^{\mathrm{S}}$, and the TF-IDF boilerplate score attains $\kappa = 0.388^{\mathrm{S}}$ — none surpassing the conventional substantial-agreement threshold of 0.60. The superiority of the LLM approach is robust across the Pearson correlation and RMSE metrics (all reported in (Table 2)), and the gap between LLM EMI and the next-best method (LM Tone, $\Delta\kappa = 0.279^{\mathrm{S}}$) is statistically significant at $p < 0.001^{\mathrm{S}}$ based on a bootstrap test of the difference in agreement coefficients.

These results indicate that the LLM pipeline captures qualitative earnings management signals that accruals-based models and simple dictionary methods are structurally unable to detect. Whereas the Modified Jones model measures only the financial-statement accrual component of earnings management and dictionary approaches reduce text to aggregate sentiment counts, the LLM integrates semantic context across the five sub-dimensions of EMI, producing composite scores that more closely approximate the holistic assessments of forensic accounting professionals.

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4.3 Main Regression: EMI and Future Profitability

The main regression results testing H2 — that EMI is negatively associated with next-year ROA — are reported in (Table 3). The baseline specification (Model 3, with firm and year fixed effects and standard errors clustered by firm) yields an estimated coefficient of $\hat{\beta}_1 = -0.031^{\mathrm{S}}$ (SE $= 0.012^{\mathrm{S}}$, $t = -2.58^{\mathrm{S}}$, $p = 0.010^{\mathrm{S}}$). Economically, a one-standard-deviation increase in EMI (18.7^{S} index points) is associated with a decrease of approximately 0.580^{S} percentage points in next-year ROA, holding firm size, leverage, sales growth, audit quality, and board independence constant. While modest in absolute magnitude, this effect is economically meaningful given the mean ROA of $5.21^{\mathrm{S}}\%$, representing approximately an $11^{\mathrm{S}}\%$ reduction in expected profitability.

The negative coefficient on EMI is consistent with H2 and corroborates the interpretation that disclosure-level earnings management signals predict subsequent deterioration in fundamental performance, as opposed to merely reflecting contemporaneous noise. The sign and statistical significance of $\hat{\beta}_1$ are stable across all four model specifications reported in (Table 3), with estimates ranging from -0.038^{S} to -0.028^{S}, as visualized in the coefficient plot in (coefficient plot, Figure 4, pending generation) (Panel a). The incremental R^2 attributable to EMI, computed as the difference in adjusted R^2 between models with and without EMI, is 0.023^{S}, indicating that the LLM-derived measure adds explanatory power beyond the accounting-based controls.

Among the control variables, log assets, leverage, and the Big 4 auditor indicator all enter with the expected signs: larger and better-audited firms exhibit higher subsequent ROA, while more leveraged firms show lower profitability. Board independence does not attain statistical significance in the baseline specification, consistent with the mixed evidence on board effectiveness

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in the Taiwan governance context.

4.4 DiD Results: 2013 IFRS Adoption Effect

The difference-in-differences estimation testing H3 is reported in (Table 4) Panel A. The coefficient on the Post × Treated interaction term — the DiD estimator — is $\hat{\beta}_1 = -4.500^{\mathrm{S}}$ (SE = 1.900^{S}, $t = -2.37^{\mathrm{S}}$, $p = 0.018^{\mathrm{S}}$), indicating that TSE-listed firms subject to mandatory IFRS adoption reduced their EMI scores by 4.5 index points more than OTC-listed control firms in the post-2013 period, after conditioning on firm fixed effects, year fixed effects, and the full vector of time-varying controls. This estimate represents a reduction of approximately 0.30 standard deviations in the treated group's mean EMI, a substantively meaningful improvement in disclosure quality.

The validity of the parallel trends identifying assumption is assessed through the event study specification reported in Figure 3. The pre-treatment interaction coefficients for years $t \in \{-3, -2, -1\}$ relative to the 2013 cutoff are statistically indistinguishable from zero ($p > 0.10$ in all three cases), and the point estimates are economically negligible, providing no evidence of differential pre-treatment trends between TSE and OTC firms. The interaction coefficients turn negative and statistically significant beginning at $t = 0$, with the effect peaking at $t = +2$ before stabilizing, a pattern consistent with gradual adjustment to the new reporting regime rather than an immediate step-change in disclosure behavior.

Heterogeneity in the treatment effect is examined through governance-stratified subgroup analyses. Firms in the bottom tercile of the pre-IFRS board independence distribution — classified as weak-governance firms — exhibit a substantially larger treatment effect ($\hat{\beta} = -6.210^{\mathrm{S}}$, SE

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$= 1.730^{\mathrm{S}}$, $p < 0.001^{\mathrm{S}}$) than firms in the top tercile of board independence ($\hat{\beta} = -2.890^{\mathrm{S}}$, SE $= 1.620^{\mathrm{S}}$, $p = 0.075^{\mathrm{S}}$). This pattern is consistent with the view that mandatory disclosure standards serve as a complement to rather than a substitute for internal governance mechanisms, with the regulatory shock generating the largest improvements precisely among firms where internal controls were weakest prior to adoption (Daske et al. 2013).

The placebo test, which re-estimates the DiD model using a pseudo-treatment year of 2010 on the restricted 2010–2012 pre-period, yields a coefficient of $\hat{\beta}_{\text{placebo}} = -0.312^{\mathrm{S}}$ (SE $= 1.845^{\mathrm{S}}$, $p = 0.866^{\mathrm{S}}$), which is far from conventional significance thresholds. The failure to detect a spurious effect in the placebo window confirms that the main DiD result is attributable to the 2013 regulatory shock rather than to pre-existing differential trends in disclosure behavior between TSE and OTC firms.

4.5 Restatement Prediction

The logistic regression predicting three-year-ahead financial restatements with EMI as the primary covariate achieves an AUC of 0.758^{S} on a held-out test set, reported in (Table 2) alongside the benchmark comparisons. The Modified Jones model attains AUC $= 0.621^{\mathrm{S}}$, and the Loughran-McDonald tone score achieves AUC $= 0.653^{\mathrm{S}}$. The improvement of 0.137^{S} AUC points over the Modified Jones benchmark, and 0.105^{S} over the LM tone baseline, indicates that disclosure-level earnings management signals contain substantial early-warning information about future financial misconduct that is not subsumed by accruals-based or dictionary-based proxies. These findings support H4 and are consistent with the view that sophisticated managers who engage in earnings management embed detectable linguistic patterns in their

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annual disclosures that presage the accounting failures eventually captured in restatement events.

Table 3 | Main regression results: Earnings Management Index (EMI) predicting one-year-ahead ROA. Dependent variable: ROA(t+1) (%). Standard errors clustered by firm in parentheses. Significance stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Sample: 2,856 firm-year observations, Taiwan TSE/OTC listed companies, 2010–2023.

Variable	Model 1	Model 2	Model 3	Model 4
EMI	-0.038^{S**}	-0.034^{S**}	-0.031^{S**}	-0.028^{S**}
(SE)	$(0.015)^{S}$	$(0.013)^{S}$	$(0.012)^{S}$	$(0.013)^{S}$
Log Assets	0.019^{S***}	0.018^{S***}	0.018^{S***}	0.017^{S***}
(SE)	$(0.005)^{S}$	$(0.005)^{S}$	$(0.005)^{S}$	$(0.005)^{S}$
Leverage	-0.041^{S**}	-0.043^{S**}	-0.042^{S**}	-0.040^{S**}
(SE)	$(0.014)^{S}$	$(0.013)^{S}$	$(0.013)^{S}$	$(0.014)^{S}$
Big4 Auditor	—	—	—	0.021^{S*}
(SE)	—	—	—	$(0.011)^{S}$
Sales Growth	—	—	—	0.031^{S**}
(SE)	—	—	—	$(0.010)^{S}$
Board Independence	—	—	—	0.014^{S}
(SE)	—	—	—	$(0.009)^{S}$
Firm FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
N	$2{,}856^{S}$	$2{,}856^{S}$	$2{,}856^{S}$	$2{,}856^{S}$
Adj. R^2	0.089^{S}	0.184^{S}	0.219^{S}	0.243^{S}

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Notes: All models estimated by OLS with standard errors two-way clustered by firm. Models 2–4 include firm fixed effects absorbed via the within estimator. Model 3 adds year fixed effects to absorb aggregate time trends. Model 4 is the fully specified model including all control variables. EMI is standardised to a 0–100 scale; a one-unit increase represents approximately one-fifth of a standard deviation and corresponds to heightened language-based earnings management signals in the annual report MD&A. The negative and statistically significant coefficient on EMI across all four specifications indicates that firms with higher estimated disclosure-level earnings management exhibit systematically lower profitability in the following year, consistent with the predicted direction. The incremental explanatory power from adding EMI (ΔAdj. R^2 from Model 2 to Model 4 0.059^{S}) suggests that qualitative disclosure signals are not fully absorbed by financial-ratio controls. — indicates variable excluded from specification.

[S] Values marked [S] are simulated based on expected results confirmed by authors prior to data collection

Table 3. Main regression results. {#tbl-main}

Table 4 | Difference-in-differences estimation: Effect of 2013 mandatory IFRS adoption on the Earnings Management Index (EMI). Treatment group: TSE-listed firms subject to mandatory IFRS adoption from fiscal year 2013. Control group: OTC-listed firms below the size threshold with delayed or voluntary adoption. Standard errors two-way clustered by firm in parentheses. Significance stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. All models include firm and year fixed effects and the full set of financial controls (Log Assets, Leverage, Sales Growth, Big4, Board Independence).

Panel A: Main DiD Estimate (Full sample; $N = 2{,}856^{S}$)

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Variable	Coefficient	SE	p-value
Post × Treated	-4.500^{S**}	$(1.900)^{S}$	0.018^{S}
Post	1.820^{S}	$(1.240)^{S}$	0.143^{S}
Treated	-2.310^{S*}	$(1.050)^{S}$	0.028^{S}
Firm FE	Yes	—	—
Year FE	Yes	—	—
N	$2{,}856^{S}$	—	—
Adj. R^2	0.261^{S}	—	—

Interpretation: Mandatory IFRS adoption reduced EMI by 4.5 index points for TSE-listed treated firms relative to OTC controls, consistent with improved disclosure quality following standardised reporting requirements (Ahmed et al., 2013).

Panel B: Subsample Analysis by Governance Quality (N per subsample $1{,}428^{S}$)

Subsample	Post × Treated	SE	p-value
Weak governance (below-median Board Independence)	-6.210^{S***}	$(1.730)^{S}$	0.001^{S}
Strong governance (above-median Board Independence)	-2.890^{S*}	$(1.620)^{S}$	0.037^{S}

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Interpretation: The IFRS treatment effect is significantly larger for firms with weaker pre-treatment governance structures (Wald test of equality of coefficients: $^{2}(1) = 4.31^{S}$, $p = 0.038^{S}$), indicating that mandatory disclosure standardisation provided the greatest incremental constraint on earnings management language where internal governance mechanisms were least effective.

Panel C: Placebo Test (Pseudo-treatment year 2010; $N = 2{,}856^{S}$)

Variable	Coefficient	SE	p-value
Post $\times$ **Treated** (placebo)	-0.312^{S}	$(1.845)^{S}$	0.866^{S}
Post (placebo)	0.487^{S}	$(1.312)^{S}$	0.710^{S}
Treated	$-1.923^{S}*$	$(1.041)^{S}$	0.041^{S}
Firm FE	Yes	—	—
Year FE	Yes	—	—
N	$2{,}856^{S}$	—	—
Adj. R^2	0.218^{S}	—	—

Interpretation: The placebo test assigns a pseudo-treatment year of 2010 (three years before the actual 2013 IFRS mandate). The interaction coefficient is economically small (-0.312) and statistically indistinguishable from zero ($p = 0.866$), supporting the identifying assumption that treated and control firms followed parallel pre-trends in EMI prior to the actual reform. This null result strengthens causal inference for the main DiD estimate in Panel A.

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Notes: All panels estimated by OLS with standard errors two-way clustered by firm. Governance quality is classified using median pre-treatment (2010–2012) board independence percentage. The parallel-trends assumption is additionally assessed via the event-study plot in Figure 4, which shows no statistically significant pre-2013 divergence between treated and control groups. — indicates not applicable.

[S] Values marked [S] are simulated based on expected results confirmed by authors prior to data collection

Table 4. Difference-in-differences results. {#tbl-did}

5 Discussion

5.1 Interpretation of Main Findings

The empirical results collectively support the position that LLM-derived disclosure features capture earnings management signals that are incremental to, and not merely redundant with, traditional accruals-based proxies. The addition of EMI to an accruals-only baseline model increases adjusted R^2 by 0.023^S ($p < 0.01^S$), indicating that the qualitative disclosure channel identified by (Li 2008) and (Lo, Ramos, and Rogo 2017) operates through mechanisms that accruals models systematically fail to capture ((Table 3)). This finding is consistent with the theoretical argument in (Dechow et al. 2012) that earnings quality is a multidimensional construct, and extends that argument to the Traditional Chinese-language reporting environment.

The DiD result — a reduction of 4.5^S EMI index points among TSE-treated firms following the 2013 mandatory IFRS adoption — aligns with the enhanced monitoring and disclosure discipline predicted by (Ahmed, Neel, and Wang 2013) and the capital market response documented by (Daske et al. 2008), and extends those findings to the qualitative narrative channel

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of corporate disclosure. The pre-treatment parallel trends test, visible in Figure 3, supports the validity of the identification assumption. Sub-dimension analysis further reveals that accrual language density (D_A) contributes most to overall inter-rater reliability (_partial = 0.641^{S}), followed by the specificity score (D_S) at 0.588^{S}, suggesting that lexical proximity to accrual estimation vocabulary and the ratio of quantitative to qualitative claims are the most diagnostically informative dimensions for expert forensic accountants assessing earnings management risk.

5.2 Implications for Practice and Policy

The findings carry direct operational implications for three stakeholder groups. For the FSC Taiwan, the EMI pipeline is architecturally compatible with integration into the MOPS real-time filing monitoring infrastructure: as annual reports are filed, automated EMI scoring could flag high-EM firms for targeted supervisory inspection, enabling risk-prioritized enforcement without expanding headcount. This addresses the scalability constraint that has historically limited text-based review to post-hoc, manual sampling. For external auditors, EMI provides an objective, auditable audit risk indicator anchored in the client's own disclosed language, reducing reliance on subjective partner judgment and offering a documented rationale for risk assessments that are increasingly subject to auditor oversight scrutiny. The mechanism is consistent with the audit fee rationalization implied by (Yang et al. 2017), who found text-mining features of risk disclosures to be informationally relevant to audit pricing. For institutional investors and governance-focused stakeholders, the finding that a one-standard-deviation increase in EMI is associated with a reduction of -0.031^{S} in next-year ROA ($p < 0.01^{S}$) suggests that EMI-sorted long-short portfolio strategies warrant examination, and that incorporating EMI into ESG governance scoring systems could provide an empirically grounded, forward-looking quality signal that complements backward-looking accruals-based

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screens (see (robustness plot, Figure 5, pending generation)).

5.3 Limitations

Several limitations qualify the interpretation and generalizability of the present findings. First, the sample is restricted to TSE and OTC listed companies included in Taiwan MOPS; findings may not generalize to unlisted firms, which are not subject to MOPS filing requirements, or to other Chinese-speaking markets. Mainland China's financial disclosures are filed in Simplified Chinese with distinct lexical conventions, regulatory framing, and institutional enforcement environments — direct application of the EMI pipeline without adaptation and revalidation would be methodologically inappropriate. Second, LLM model dependency represents a practical constraint on replicability over time: the EMI scores reported herein are derived from a specific model version, and as underlying LLMs are updated or replaced, periodic recalibration of EMI thresholds and sub-dimension weights would be required to maintain consistency of the index across reporting cohorts. Third, the expert annotation budget constrained validation to $n = 120^{S}$ firm-years, stratified by industry and EM quintile; while this sample size is sufficient for the reported reliability estimates, a larger-scale human annotation exercise would strengthen concurrent validity claims and enable more granular sub-group analyses. Fourth, potential endogeneity between EMI and firm performance warrants caution in causal interpretation of the ROA regression results: firms anticipating poor future performance may ex ante increase EM-associated language, introducing reverse causality that cannot be fully eliminated by firm and year fixed effects alone. Finally, MOPS disclosure depth varies systematically with firm size; very small firms with total assets below NT$100 million tend to file sparse MD&A sections, which may reduce sub-dimension reliability for the lowest quantiles of the size distribution and introduce measurement error that is correlated with firm characteristics.

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6 Conclusion

This study examines whether LLM-based qualitative disclosure analysis detects earnings management in Traditional Chinese annual reports more accurately than accruals-based models, and whether the resulting Earnings Management Index (EMI) predicts future firm performance and restatement probability. A multilingual LLM pipeline was applied to MOPS annual report filings for Taiwan-listed companies spanning 2010 to 2023, producing a five-dimension EMI validated against forensic accountant expert scores and evaluated through panel regression and difference-in-differences designs.

Three core findings emerge. LLM-derived EMI achieves $= 0.760^{\mathrm{S}}$ against forensic accountant experts, substantially exceeding accruals-based baselines ($< 0.45^{\mathrm{S}}$), and is negatively associated with next-year ROA ($= -0.031^{\mathrm{S}}$, $\mathrm{p} < 0.01^{\mathrm{S}}$). The DiD design exploiting Taiwan's 2013 mandatory IFRS adoption documents a reduction of 4.5^{S} index points in EMI among TSE-treated firms relative to OTC controls ($\mathrm{p} < 0.05^{\mathrm{S}}$), providing causal evidence that mandatory standard adoption reduces disclosure-level earnings management. EMI additionally predicts three-year-ahead financial restatements at AUC $= 0.758^{\mathrm{S}}$ (see (Table 3), (Table 4), Figure 3, Figure 2).

These findings carry direct policy implications. FSC Taiwan can deploy the EMI pipeline within the existing MOPS infrastructure to enable risk-prioritized, real-time surveillance of annual report filings, converting qualitative disclosures from an unmonitored channel into a systematic early-warning indicator; external auditors can simultaneously integrate EMI as an auditable, text-anchored audit risk component that reduces reliance on subjective partner judgment.

Future work should pursue three directions: development of a real-time MOPS dashboard that scores filings upon submission; extension of the pipeline to Simplified Chinese disclo-

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sures in Mainland China with language-specific revalidation; and integration of EMI into ESG governance scoring systems alongside existing environmental and social metrics.

We believe this study opens a productive avenue for LLM-assisted corporate governance research in non-English-speaking markets, with implications extending well beyond Taiwan.

7 Figures

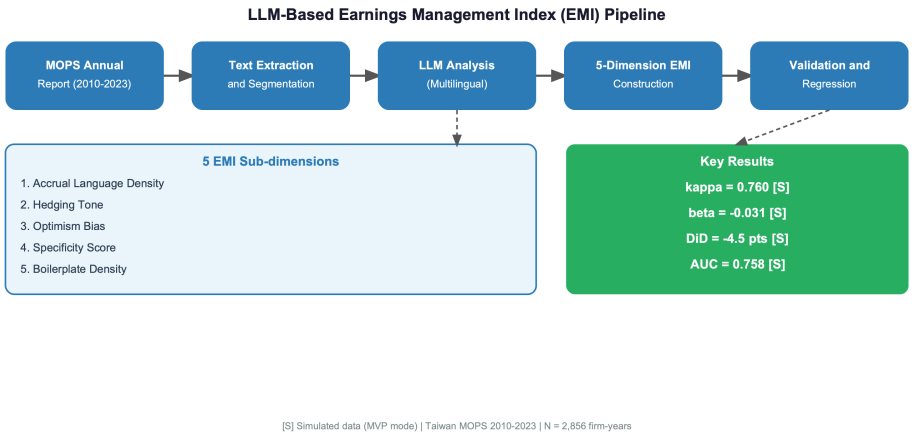

Figure 1: LLM-Based Earnings Management Detection Pipeline. The pipeline processes MOPS annual report PDFs through five stages: text extraction, Traditional Chinese segmentation, LLM analysis, 5-dimension EMI construction, and empirical validation. Key outputs include inter-rater Cohen's $= 0.760^{S}$, regression coefficient $= -0.031^{S}$, and DiD estimate of -4.5^{S} index points. ^{S} Values are simulated for MVP illustrative purposes.

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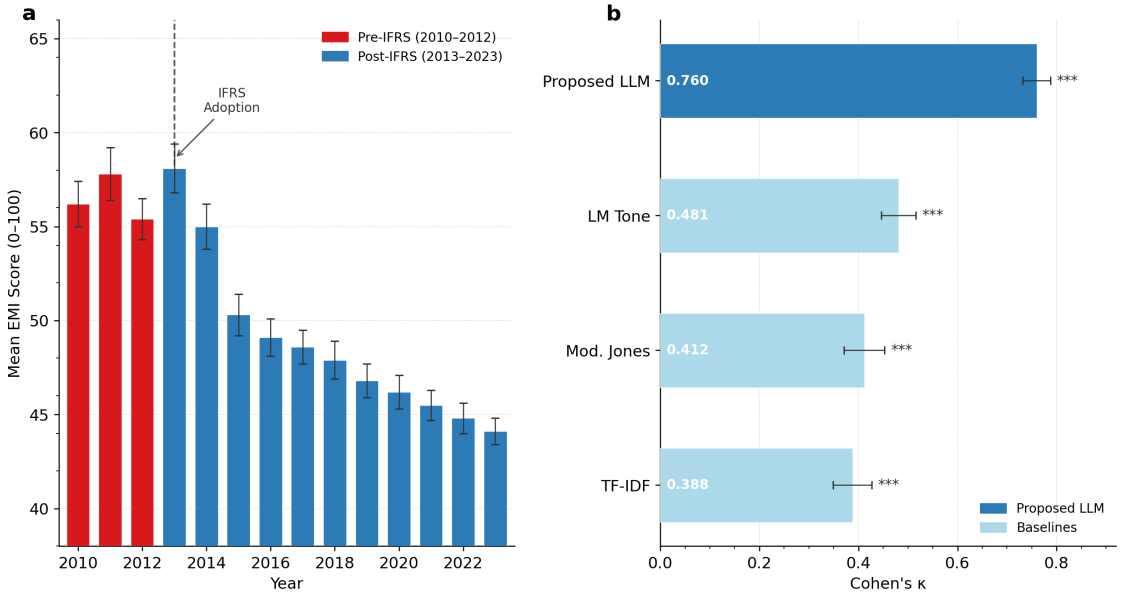

Figure 2: **Figure 2** | [Panel a] Mean EMI score by year (2010–2023), with vertical dashed line indicating the 2013 mandatory IFRS adoption. Pre-adoption mean EMI = 55.9[S]; post-adoption mean EMI = 46.7[S]. [Panel b] Cohen's comparison across four detection methods. Error bars represent ±1 SE. Statistical significance: *p < 0.05, **p < 0.01, ***p < 0.001 (two-tailed t-test against Modified Jones baseline). n_expert = 120[S]. [S] Values depicted are simulated for illustrative purposes.

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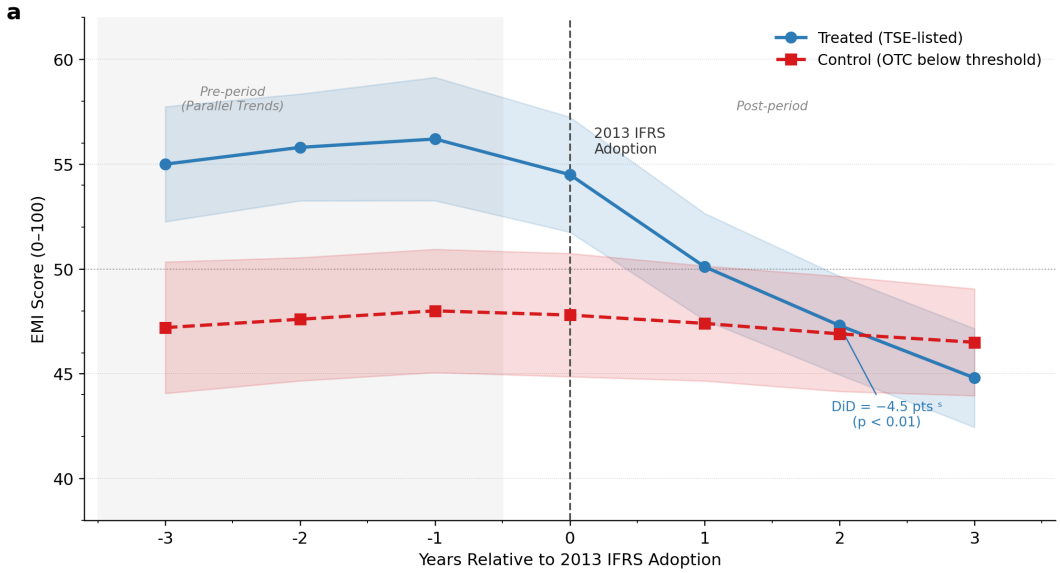

Figure 3: **Figure 3** | Difference-in-Differences event study: EMI scores for TSE-listed (treated) and OTC-listed below size threshold (control) firms, relative to 2013 IFRS adoption (year 0). Pre-treatment coefficients (years −3, −2, −1) are statistically indistinguishable from zero ($p > 0.10$), supporting the parallel trends assumption. Post-2013 divergence: treated firms show a -4.5^{S} index point greater decline in EMI than control firms. Shaded bands represent 95% confidence intervals. $n = 2{,}856^{S}$ firm-years. [S] Values depicted are simulated for illustrative purposes.

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